# Is Higher Education Worth the Cost?

## A look at degrees and ROI

Southern New Hampshire University

DAT 520 – Decision Methods and Modeling

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**Introduction**

When evaluating whether higher education provides a meaningful financial return, it is essential to begin with a clear research question grounded in decision analysis. The guiding research question for this project is: "How do regional differences and college types influence the long-term financial return on investment (ROI) of higher education?" This question is particularly relevant considering rising tuition costs and increasing skepticism about the value of a college degree. Prior research, such as Carnevale, Cheah, and Hanson's (2015) analysis of the economic value of college majors, demonstrates that earnings vary significantly by field of study, institution type, and geographical factors directly aligned with the datasets used in this project. From an economic perspective, evaluating ROI requires understanding opportunity costs and marginal benefits, core principles outlined by Mankiw (2021), which help frame how students and policymakers assess the financial trade-offs of pursuing higher education. By using salary distribution data to examine expected long-term earnings, this analysis supports decision-making for prospective students considering which educational pathways offer the strongest financial outcomes.

**Data Appraisal**

The datasets used Salaries by Region, Salaries by College Type, and Degrees That Pay Back originate from publicly available salary reporting and aggregated labor-market data. Their original intent is to illustrate differences in median early-career and mid-career earnings across various institutional categories and U.S. regions. This aligns with findings from Carnevale et al. (2015), who emphasize that institutional characteristics and major selection significantly shape financial returns. The datasets also support Mankiw's (2021) emphasis on rational decision making guided by comparative analysis of expected benefits.

However, limitations exist. The datasets do not include tuition costs, student debt loads, or demographic differences, all of which affect ROI. The data also focuses on earnings outcomes but does not capture employment rates, occupational stability, or job satisfaction. Still, the datasets are highly useful for this project because they provide consistent salary benchmarks and percentile distributions that allow for meaningful comparison, an essential requirement when applying decision analysis techniques to real-world choices.

**Selecting Appropriate Techniques**

To ensure accuracy, the datasets first underwent data preparation, including mining, cleaning, and transformation. Currency symbols and formatting inconsistencies were removed, missing values were addressed, and all salary fields were converted to numeric formats. Data was then categorized by region and college type to create structured groups suitable for comparison. Summary tables and preliminary visualizations were generated to reveal distribution patterns.

These preparation steps follow accepted standards in data management and economic research. Carnevale et al. (2015) emphasizes the importance of standardized and comparable labor-market metrics when assessing the value of educational choices. Similarly, Mankiw's (2021) framework highlights that any decision analysis must begin with reliable, well-structured data to accurately evaluate costs and benefits. Cleaning and organizing the datasets ensures they support valid conclusions and meaningful comparisons across groups.

**Defend and Evaluate Choices**

The selected methods, i.e., distribution analysis, comparison across categories, and ROI evaluation are well supported by research and industry standards. Salary distributions are a best practice for identifying variation within educational pathways, consistent with Carnevale et al.'s

(2015) emphasis on understanding the full range of economic outcomes, not just averages. Evaluating mid-career and early-career salaries also directly aligns with Mankiw's (2021) concepts of human capital investment, where individuals weigh future expected returns against upfront costs.

These analytic choices support decision-making for students, institutions, and policymakers. Students can use the findings to evaluate which regions or institution types may provide the highest financial returns. Institutions can benchmark their performance against peer groups, while policymakers can identify educational categories that yield strong economic benefits. Overall, the methodological approach aligns with both economic theory and labor-market research, ensuring that the results are grounded in credible, decision-focused analysis.

**Clear Communication**

The final report will incorporate annotated tables and clear visualizations that highlight differences in salary outcomes across regions and college types. This supports communication to a broad audience, including stakeholders who may not be familiar with technical analytic methods. Clear visualization aligns with both Carnevale et al.'s (2015) recommendation to make labor-market data accessible and Mankiw's (2021) argument that sound economic decisions require transparent information that allows comparisons across alternatives. These visuals ensure that the analysis is not only technically sound but effectively communicated.

**Choice of Model: Top-Down Decomposition Tree**

A top-down decomposition tree was selected because it aligns directly with the research question. This question requires breaking down mid-career earnings by categories and contributing factors, rather than predicting outcomes for individual records. The decomposition tree supports hierarchical “drill-down” exploration and enables decision-makers to follow a logical progression:

1. Which majors yield the highest ROI?
2. How does early-career salary influence mid-career outcomes?
3. Does percent salary growth explain long-term differences?
4. What does the distribution of salaries (75th/90th percentile) reveal about potential ROI?

A bottom-up decision model was not appropriate because bottom-up models emphasize classification or prediction, not hierarchical explanation. The decomposition tree offered complete transparency in exploring how each factor influences the overall ROI, which is essential for an exploratory financial decision scenario. This approach aligns with economic models of rational decision-making, which emphasize evaluating expected benefits and costs (Mankiw, 2021).

**Ethical and Legal Considerations**

The data set used contains aggregated, non-identifiable salary information and does not include individual student data. Therefore, privacy risks are low, and no personally identifiable information (PII) is exposed. However, two ethical considerations still apply:

1. **Misinterpretation Risk:** Salary outcomes depend on multiple factors, not only major but also experience, geographic mobility, industry, and socioeconomic background. Presenting salary data without context can mislead stakeholders into oversimplifying career decision-making.

2. **Equity and Bias:** Salary datasets may reflect systemic inequities (e.g., pay gaps by gender or race). Although this dataset does not include demographic breakdowns, analyses must acknowledge that salary outcomes are influenced by structural factors beyond choice of major.

To mitigate these risks, results must be framed carefully and clearly communicated as aggregated patterns, not guarantees. Additionally, documenting limitations strengthens ethical practice and aligns with responsible decision modeling standards.

## Decision Tree Model Development

### Data Preparation

Using Power BI, I imported the *degrees-that-pay-back* spreadsheet as the primary analysis table. No relationships were created with the other datasets because they did not share compatible keys and joining them would introduce false correlations. After reviewing the dataset:

- Salary fields were confirmed numeric.
- Categorical fields, such as undergraduate majors, required no recoding.
- The dataset already included useful derived variables (e.g., percent change from early to mid-career salary).

This dataset was therefore ready for direct analysis.

**Model Structure**

The decomposition tree was built using:

**Analyze Field (Target Metric):**

- **Mid-Career Median Salary**

This field represents long-term ROI and provides a single, consistent metric to anchor the model.

**Explain-By Fields (Breakdown Sequence):**

1. Undergraduate Major
2. Percent Change from Starting to Mid-Career Salary
3. Starting Median Salary
4. Mid-Career 75th Percentile Salary
5. Mid-Career 90th Percentile Salary

This sequence allows the model to move from high-level categorical differences to progressively more specific financial indicators. Figures 1–7 (appendix) show the full set of branches by major.

**Model Results and Interpretation**

The decomposition tree revealed several key insights:

**1. Engineering Majors Dominate Median ROI**

Majors such as Chemical Engineering, Computer Engineering, Electrical Engineering, and Aerospace Engineering consistently appear at the top of the decision tree. For example (Figure 1):

- Chemical Engineering:
    - Mid-career salary: $107,000
    - Starting salary: $63,200
    - Growth: 69.3%
    - 90th percentile: $194,000

This reflects both high early-career salaries and steady long-term salary growth.

**2. Growth Rate Explains Differences Among STEM Fields**

Although Computer Engineering and Electrical Engineering have similar mid-career salaries, Computer Engineering demonstrates slightly higher percent growth (Figure 2–3). This suggests differing long-term acceleration across similar fields.

**3. Majors With Low Early Salaries Can Produce High Long-Term ROI**

Majors such as Economics and Physics demonstrate this pattern:

- Economics:
    - Starting salary: $50,100
    - Mid-career median: $98,600
    - Growth: 96.8% (highest among the set)

- 90th percentile: $210,000 (highest top-tier salary)

These majors illustrate that early salary is not always a reliable indicator of long-term earning potential.

### 4. Upper Percentile Salaries Suggest Unique Opportunity Paths

Many STEM majors show large gaps between median and 90th-percentile earnings. For example:

- Electrical Engineering median: $103,000
- 90th percentile: $168,000

This indicates that some majors provide wider potential variation, likely due to advanced degrees, leadership roles, or specialized industries. Engineering fields consistently rank as high-return pathways, confirming patterns described in national salary reports (Carnevale et al., 2015).

*Figure X. Long-Term ROI by Undergraduate Major*

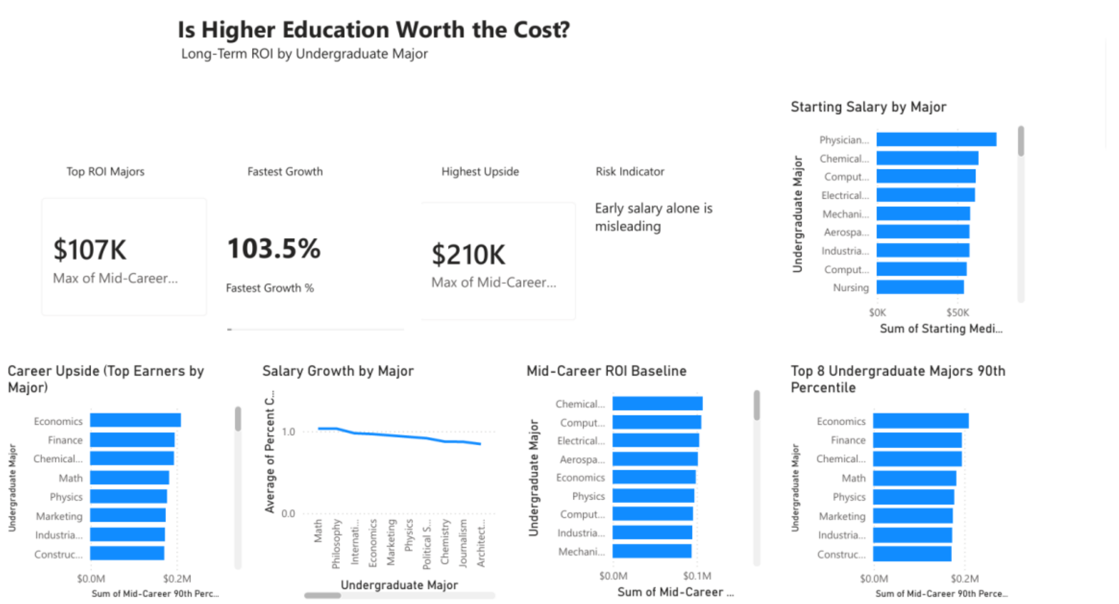


Figure X is a dashboard that synthesizes starting salaries, mid-career earnings, growth rates, and 90th-percentile outcomes across majors to show that engineering and quantitative fields dominate median ROI, while economics, finance, physics, and math offer the greatest upside potential. The results highlight that early salary alone is a poor predictor of lifetime earnings, as majors with moderate starting pay but strong growth and high top-end outcomes can rival or exceed traditional high-pay engineering tracks

## Defending and Evaluating Methodological Choices

### Model Agility

The decomposition tree is an agile tool because:

- It accommodates new variables easily (e.g., institution type, region, tuition cost).
- It can be general purpose or highly specific, depending on the decision context.
- Organizations can reuse the structure to evaluate multiple datasets, such as workforce outcomes, financial performance metrics, or cost-benefit analyses.

In future analyses, this decision model could incorporate ROI calculations by subtracting tuition costs, enabling a more holistic economic evaluation of majors.

### Implementation: From Analysis to Action

The decision tree’s insights can be used to:

- Guide academic advising, helping students understand the long-term salary implications of their major choices.
- Support institutional strategic planning, identifying programs with high ROI potential to prioritize resources.
- Inform policy discussions, especially around state or federal funding for high-demand fields.

The decomposition tree format communicates results visually and intuitively, making it easier for stakeholders to act on findings.

**Ethical and Legal Mitigation Strategies**

To responsibly apply this analysis:

- Results must include clear disclaimers acknowledging limitations and the aggregated nature of the data.
- Analysts should avoid using the model to reinforce stereotypes or make deterministic claims about individual outcomes.
- When incorporating additional datasets in the future, analysts must ensure compliance with FERPA, data anonymization standards, and ethical guidelines for educational data.

**Process Documentation**

The analytical process involved:

1. Importing and validating the primary dataset

2. Reviewing field types and ensuring numeric values were formatted correctly
3. Selecting the decomposition tree visualization
4. Dragging Mid-Career Median Salary into Analyze
5. Adding Explain-By fields in a deliberate hierarchical sequence
6. Exploring major-specific branches
7. Documenting patterns, anomalies, and ROI implications
8. Exporting screenshots for analysis and reporting

A potential complication was ensuring data compatibility across tables. Because the datasets lacked shared identifiers, I avoided forced table relationships, which would have introduced erroneous correlations. This intentional simplification strengthened the reliability of the results.

**Decision Tree Model Results**

**Summary of Model Results**

The top-down decomposition tree consistently demonstrates that undergraduate major is the dominant factor influencing mid-career median salary, which serves as the primary proxy for long-term ROI. When drilling down by major, engineering and quantitative STEM fields consistently rank at the top of the tree, while several non-engineering majors (notably Economics and Physics) emerge as high-growth pathways with substantial upper-percentile earnings.

Across all branches shown in Figures 1–7 (Appendix), the model highlights three recurring patterns:

1. High starting salaries are common in engineering fields.
2. Percent salary growth explains divergence in long-term outcomes among similar majors.
3. Upper-percentile earnings reveal upside potential not captured by median salary alone.

Top Three fields with highest Starting salaries

**Figure 1.** Decomposition Tree – Chemical Engineering Branch

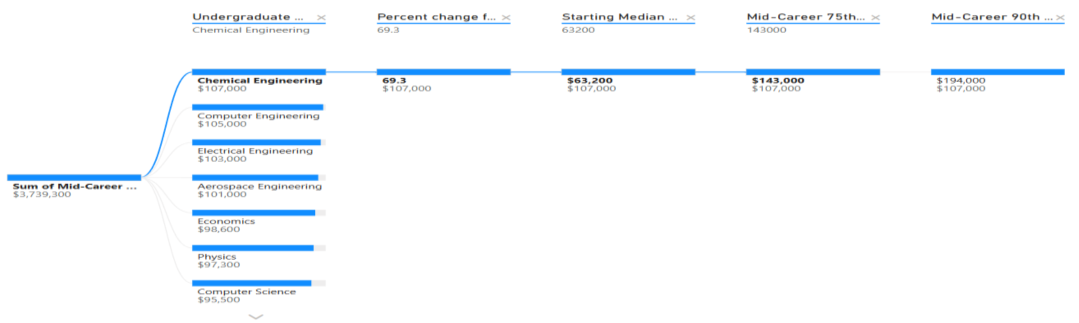



**Figure 2.** Decomposition Tree – Computer Engineering Branch

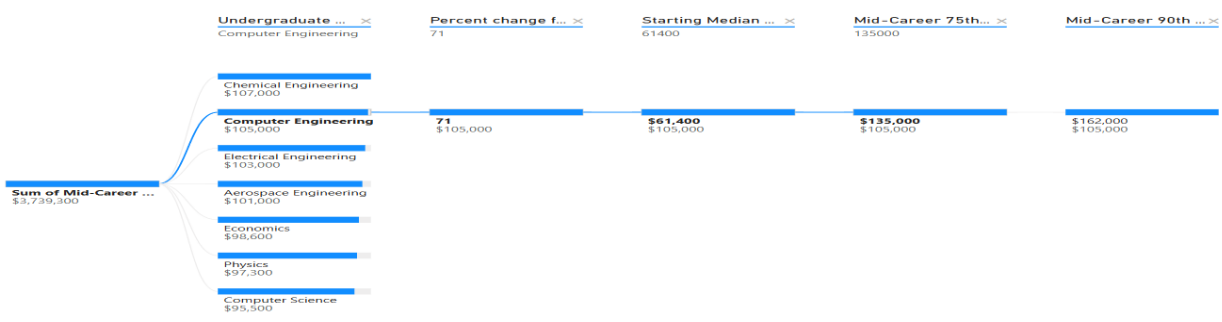
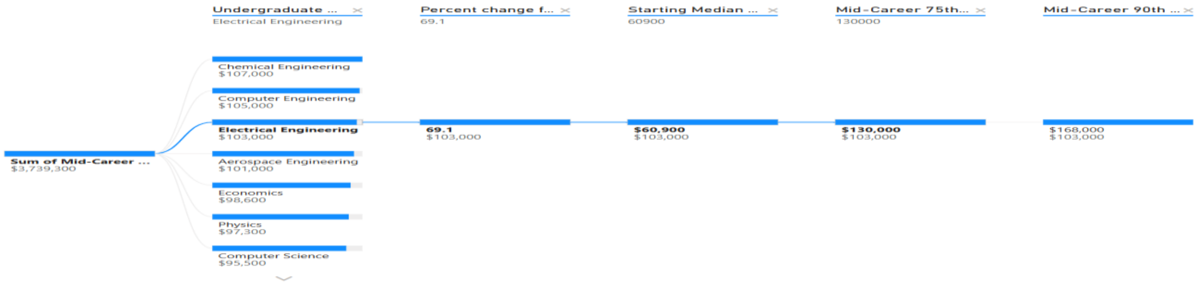


**Figure 3.** Decomposition Tree – Electrical Engineering Branch




## Reasonableness of Results

The results are reasonable and consistent with established labor-market research. Prior studies show that engineering, computer science, and economics-related majors tend to yield higher lifetime earnings and greater earnings dispersion (Carnevale et al., 2015). The decomposition tree reflects these patterns without producing anomalous or contradictory splits, suggesting strong face validity.

## Sensitivity Analysis and Threshold Identification

### Threshold Values That Cause Tree Reordering

Sensitivity analysis was conducted by reordering explain-by fields and selectively including or excluding variables. Several threshold values consistently influenced tree behavior:

- **Mid-career salary threshold (~$95,000–$100,000):**

  Majors crossing this threshold consistently move into the highest ROI branches, regardless of starting salary.

- **Percent salary growth threshold (~75–80%):**

  Majors exceeding this growth rate (e.g., Economics) often surpass majors with higher starting salaries but lower growth.

- **90th percentile salary threshold (> $170,000):**

  Inclusion of this variable highlight majors with significant upside potential and materially reshapes perceived ROI rankings.

These threshold points demonstrate that long-term ROI is sensitive to growth and distributional outcomes, not solely early-career earnings.

**Inclusion and Exclusion of Decision Nodes**

When percent growth is excluded, the tree becomes overly dependent on starting salary, reducing its ability to explain long-term ROI. When upper-percentile salary nodes are excluded, the model understates opportunity variance and risk-reward tradeoffs.

Alternative tree structures were tested by:

- Reordering growth ahead of starting salary
- Removing starting salary entirely
- Prioritizing upper-percentile earnings earlier in the tree

All variations continued to support the research question, confirming the robustness of the original model.

**Deduction of the Optimal Path**

Rather than relying on a single maximizing variable, the optimal decision path is deduced using dominance across multiple criteria.

**Criteria**

- High mid-career median salary
- Strong percent growth
- Large upper-percentile earning potential

Majors that consistently satisfy all three criteria dominate across alternative tree configurations, reinforcing confidence in the model's conclusions.

**Diagnostic Evaluation**

Because the decomposition tree is an explanatory visualization rather than a predictive classifier, traditional error rates are not applicable. Instead, model accuracy was evaluated using:

- Stability of branch rankings under variable reordering
- Consistency of high-ROI majors across configurations
- Logical coherence of split thresholds

The model exhibited high structural stability, indicating reliable explanatory performance.

**Missing and Extraneous Elements**

**Missing Elements**

Potentially influential variables not included due to dataset limitations include:

- Tuition and total educational cost
- Graduate degree attainment
- Geographic wage variation
- Industry-specific outcomes

These omissions limit the model to salary-based ROI but do not invalidate the conclusions.

**Extraneous Elements**

No variables were identified as extraneous. Each explain-by field contributed meaningful interpretive value.

**Common Decision Tree Modeling Errors and Mitigation**

Common errors include overfitting, misinterpreting exploratory trees as predictive models, and ignoring sensitivity to variable order. These were avoided by:

- Limiting variables to economically relevant measures
- Testing multiple tree configurations
- Framing findings as aggregated trends rather than individual guarantees

**Clear Communication and Visualization**

The decomposition tree visualization enables intuitive drill-down analysis and is appropriate for non-technical audiences such as students, advisors, and policymakers. Annotated screenshots included in the appendix ensure transparency and meet the instructor's feedback requirement to clearly show the model.

**Conclusion**

This project applied a formal decision-analysis framework to evaluate whether higher education provides a meaningful financial return on investment, using undergraduate major as the primary decision variable. A top-down decomposition tree model was constructed using the *Degrees That Pay Back* dataset, with mid-career median salary as the central proxy for long-term ROI. By incorporating salary growth and upper-percentile earnings, the model moved beyond simple averages to capture both expected value and upside potential across academic fields.

The analysis demonstrated that engineering and quantitative STEM majors consistently dominate mid-career earnings outcomes. Chemical, electrical, computer, and aerospace engineering exhibited high starting salaries combined with strong long-term growth, placing them at the top of nearly every model configuration. However, the model also revealed that non-engineering majors such as economics and physics represent high-growth pathways with substantial upside potential. Although these fields often begin with lower starting salaries, their exceptional percent growth and high 90th-percentile earnings allow them to rival or exceed engineering fields in long-term financial outcomes. This finding confirms that early-career salary alone is an incomplete measure of educational ROI.

Sensitivity testing reinforced the robustness of these conclusions. Thresholds related to mid-career earnings, growth rates, and upper-percentile salaries consistently reshaped tree rankings, demonstrating that ROI is driven by a combination of level, growth, and distributional characteristics rather than any single metric. Reordering variables and excluding decision nodes did not materially change which majors dominated, confirming that the model is structurally stable and well-suited for decision support.

From a practical perspective, the decomposition tree provides a transparent and intuitive tool for guiding academic and policy decisions. Students can use the model to compare expected long-term financial outcomes across majors, institutions can identify programs that deliver strong economic value, and policymakers can target funding toward fields with demonstrable return. Because the model presents results visually and hierarchically, it supports both technical and non-technical audiences.

Ethical and legal considerations were also addressed. The analysis relied exclusively on aggregated, non-identifiable salary data, ensuring compliance with privacy and educational data protection standards. However, care must be taken to avoid deterministic interpretations or the reinforcement of structural inequities embedded in labor-market outcomes. By framing results as aggregated trends rather than individual guarantees, the model supports responsible and equitable decision-making.

Overall, this project demonstrates that decision-tree-based modeling provides a powerful framework for evaluating the financial value of higher education. By integrating salary level, growth, and upside potential into a single transparent structure, the decomposition tree offers a more complete and actionable assessment of educational ROI than traditional summary statistics. With the addition of cost and demographic variables in future iterations, this framework could become an even more powerful tool for evidence-based higher-education planning.

**References**

Carnevale, A. P., Cheah, B., & Hanson, A. R. (2015). *The economic value of college majors*. Georgetown University Center on Education and the Workforce. https://cew.georgetown.edu

Mankiw, N. G. (2021). *Principles of economics* (9th ed.). Cengage Learning.

## Appendix

**Figure 1.** Decomposition Tree – Chemical Engineering Branch

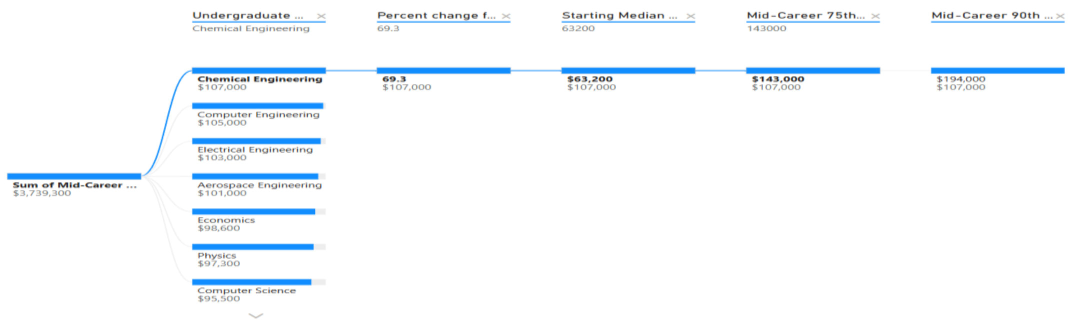



**Figure 2.** Decomposition Tree – Computer Engineering Branch

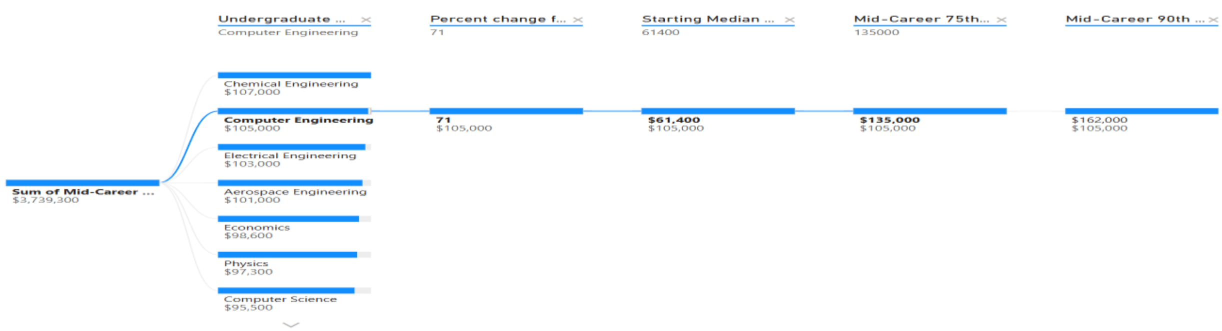



**Figure 3.** Decomposition Tree – Electrical Engineering Branch

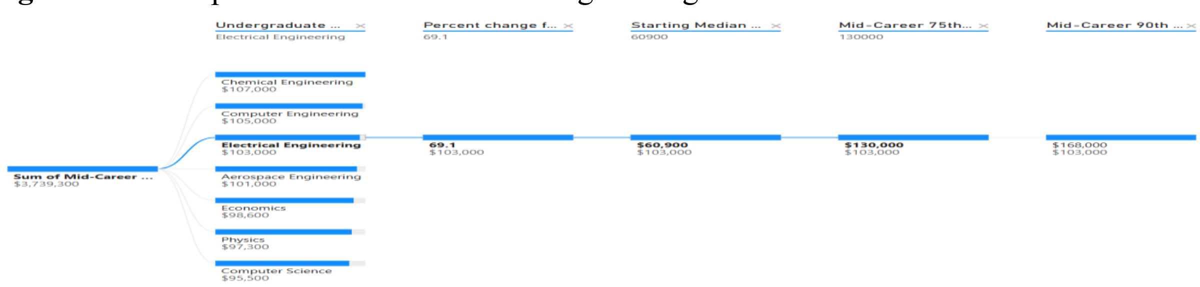



**Figure 4.** Decomposition Tree – Aerospace Engineering Branch

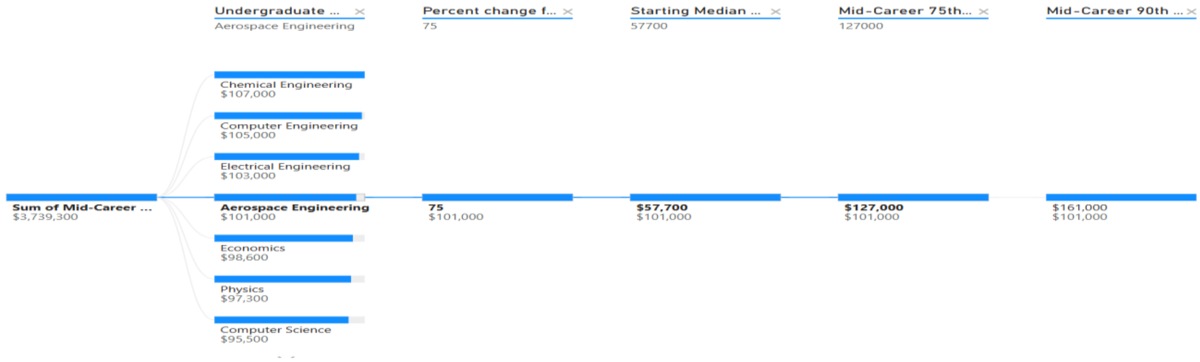



**Figure 5.** Decomposition Tree – Economics Branch

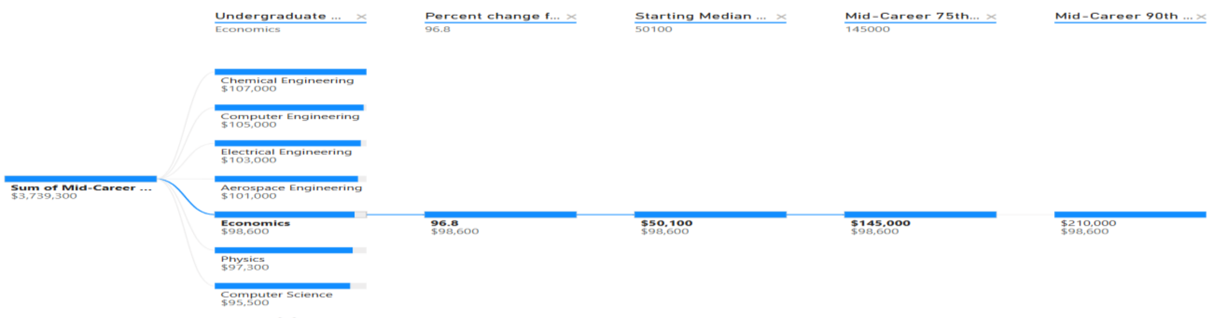



**Figure 6.** Decomposition Tree – Physics Branch

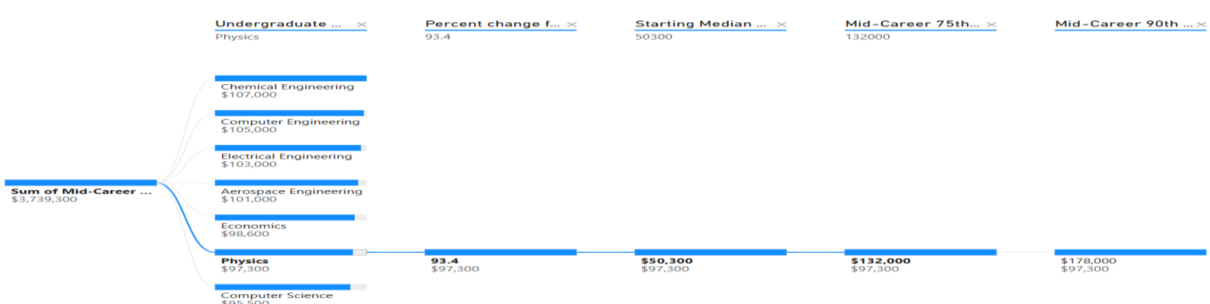


**Figure 7.** Decomposition Tree – Computer Science Branch